

Table B.2. ASL-LDAM and SimAM-DCFR implementation constants

Component	Recorded value	Role
SimAM energy constant	$\lambda = 10^{-4}$	Stabilizes the variance-dependent inverse-energy computation.
Texture path	Depthwise 3×3 , groups = C , no bias; pointwise 1×1 , no bias	Project-created local texture recalibration with preserved tensor shape.
Channel gate	GAP; 1×1 convolution, $C \rightarrow C$; sigmoid	Trainable channel-wise residual gate.
Fusion	$F' = F + \frac{1}{2}G \odot (F_{\text{SimAM}} + F_{\text{texture}})$	Residual integration of SimAM and project-created texture features.
ASL focusing	$\gamma_{\text{pos}} = 0, \gamma_{\text{neg}} = 4$	Leaves the true-class term unfocused and suppresses easy non-target terms.
LDAM	$m_{\text{max}} = 0.5$, scale $s = 30$	Applies a class-dependent margin to the true-class logit before softmax.
Insertion contract	$F' \in \mathbb{R}^{B \times C \times H \times W}$ before the original classification head	Preserves feature shape and four-logit output contract.

Only the SimAM path is parameter-free. The texture path and channel gate contain trainable convolutions. At inference, class selection uses the exported four-score output and deterministic class mapping; ASL-LDAM margins and loss weights are absent from the mobile graph.

Appendix C - Classification Candidate Inventory

This appendix preserves the screened model identities without reconstructing missing performance rows. The deployment-oriented preference is approximately 15M parameters or less; larger candidates remain useful diagnostic references when their evidence was retained.

Table C.1. TIMM candidate inventory

Candidate	Family	Retained evidence status	Deployment status
convnext_tiny_in22k	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobilenet_v3_large	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
efficientnet_b0	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
repvgg_a0	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate

Candidate	Family	Retained evidence status	Deployment status
efficientnet_v2_s	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
shufflenet_v2_x1_0	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
fastvit_t8	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
edgenext_xx_small	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobileone_s0	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobilevit_s	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobilenetv4_conv_small	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mnasnet_100	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
ghostnetv2_100	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
rexnet_100	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
squeezenet1_1	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobilenetv4_hybrid_medium	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
efficientvit_m1	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate

Table C.2. Ultralytics classification candidate inventory

Candidate	Family	Retained evidence status	Deployment interpretation
yolo26m-cls	YOLO26	Complete representative performance and efficiency row retained	Eligible under the final deployment-oriented preference
yolo26x-cls	YOLO26	Complete representative performance and efficiency row retained	Higher-capacity diagnostic control; above the approximately 15M preference
yolov8m-cls	YOLOv8	Complete representative performance and efficiency row retained	Above the approximately 15M preference in the retained candidate screen
yolov8s-cls	YOLOv8	Complete representative performance and efficiency row retained	Eligible by parameter preference
yolo11m-cls	YOLO11	Complete representative performance and efficiency row retained	Eligible by parameter preference
yolov8x-cls	YOLOv8	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolov8l-cls	YOLOv8	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo26l-cls	YOLO26	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo26n-cls	YOLO26	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo11x-cls	YOLO11	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo11s-cls	YOLO11	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo11n-cls	YOLO11	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolov8n-cls	YOLOv8	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo26s-cls	YOLO26	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo11l-cls	YOLO11	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed

The inventory records screening provenance rather than a complete ranking. A missing metric row is not reconstructed from architecture name, parameter family, or later model performance.